MATH 241: Calculus III

Fall 2026

Lecture 14: Extrema and Critical Points

Background & Motivation

In Calc I, you found maxima and minima by setting $f'(x) = 0$. The same idea works here: set $\nabla f = \mathbf{0}$. But in multiple dimensions, something new appears—saddle points, where the surface goes up in one direction and down in another. Today we develop the second derivative test for classifying critical points, then tackle global extrema on closed regions by analyzing boundaries.

1. Critical Points in Several Variables

Definition 1: Critical Point

A point (a, b) is a **critical point** of $f(x, y)$ if:

$$\nabla f(a, b) = \mathbf{0} \quad \text{i.e.,} \quad \underline{\hspace{6cm}}$$

or if ∇f is undefined at (a, b).

– All local extrema occur at critical points (necessary condition)
– But not every critical point is an extremum (could be a saddle)

2. The Second Derivative Test

Theorem 1: Second Derivative Test for $f(x, y)$

At a critical point (a, b), compute the **discriminant**:

$$\boxed{D = \underline{\hspace{6cm}}}$$

Classification:

- $D > 0$ and $f_{xx} > 0$: ________________
- $D > 0$ and $f_{xx} < 0$: ________________
- $D < 0$: ________________
- $D = 0$: ________________

The Hessian Matrix

The discriminant D is the determinant of the **Hessian**:

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}, \qquad D = \det(H) = f_{xx}f_{yy} - (f_{xy})^2$$

– $D > 0$ means both eigenvalues of H have the same sign (bowl shape)
– $D < 0$ means eigenvalues have opposite signs (saddle shape)

Example 1: $f(x, y) = x^3 + y^3 - 3xy$

Find and classify all critical points.
1. **Find critical points:**

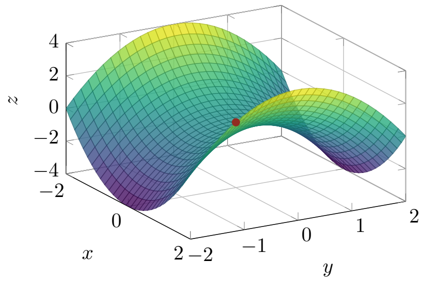
2. **Second partials:** $f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3$
3. **Classify:**
At $(0, 0)$**:** $D = (0)(0) - 9 =$ ________________ $\implies$ ________________
At $(1, 1)$**:** $D = (6)(6) - 9 =$ ________________, $f_{xx} = 6 > 0 \implies$ ________________, $f(1, 1) = -1$

Example 2: $f(x, y) = 2x^2 + y^2 - 4x - 2y + 6$

Find and classify all critical points.
1. **Critical points:**
$f_x = 4x - 4 = 0 \implies x =$ ________________
$f_y = 2y - 2 = 0 \implies y =$ ________________
One critical point: $(1, 1)$.
2. **Second partials:** $f_{xx} = 4, \quad f_{yy} = 2, \quad f_{xy} = 0$
3. **Classify:** $D = (4)(2) - 0 = 8 > 0, \quad f_{xx} = 4 > 0$
$\implies$ **Local minimum** at $(1, 1)$, $\quad f(1, 1) = 2 + 1 - 4 - 2 + 6 =$ ________________
Also the global minimum since $f \to \infty$ as $\|(x, y)\| \to \infty$.

3. Saddle Points

– A **saddle point** is a critical point that is neither a max nor a min
– The surface curves up in one direction and down in another



A saddle point at $(0,0)$ for $z = x^2 - y^2$. Along the x-axis it looks like a minimum; along the y-axis it looks like a maximum.

Example 3: The Classic Saddle: $z = x^2 - y^2$

$\nabla f = \langle 2x, -2y \rangle = \mathbf{0}$ at $(0,0)$.
$f_{xx} = 2, \quad f_{yy} = -2, \quad f_{xy} = 0. \quad D = (2)(-2) - 0 = -4 < 0.$ **Saddle point.**

4. Global Extrema on Closed Bounded Regions

Theorem 2: Extreme Value Theorem (Multivariate)

If f is continuous on a **closed, bounded** region D, then f attains both a global maximum and a global minimum on D.

Strategy for Finding Global Extrema

1. Find all critical points **inside** D (set $\nabla f = \mathbf{0}$)
2. Parameterize each piece of the **boundary**, reduce to 1D optimization on each edge
3. Evaluate f at all critical points (interior and boundary) and at all **corners**
4. Compare all values: largest = global max, smallest = global min

Example 4: Global Extrema on a Rectangle

Find the global max and min of $f(x,y) = x^2 + y^2 - 4x - 6y$ on $R = [0,4] \times [0,5]$.
1. **Interior critical points:**

2. **Boundary analysis (4 edges):**

3. **Corners:**
$f(0,0) = 0, \quad f(4,0) = 0, \quad f(0,5) = -5, \quad f(4,5) = -5$
4. **Compare all values:** $\{-13, -4, -9, -9, -9, 0, 0, -5, -5\}$

$$\boxed{\text{Global max} = 0 \text{ at } (0,0) \text{ and } (4,0). \quad \text{Global min} = -13 \text{ at } (2,3).}$$

Practice Problems

Problem 1. Find and classify all critical points of $f(x, y) = x^2 + y^2 + 2x - 4y + 5$.

Problem 2. Find and classify all critical points of $f(x, y) = x^2y - x^2 - 2y^2$.

Problem 3. Show that $f(x, y) = x^2 - y^2$ has a saddle point at the origin. Explain geometrically why it is neither a max nor a min.

Problem 4. Find the global extrema of $f(x, y) = x^3 - 3x + y^2$ on $R = [-2, 2] \times [-1, 1]$.

Problem 5. Find and classify all critical points of $f(x, y) = xy(1 - x - y)$.